

Technology Stack

Date	08 October 2026
Team ID	UNM1611Firefox
Project Name	AI StudyBuddy API
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	Used to build the user interface for uploading study materials and viewing summaries, flashcards, quizzes, and study plans. Provides a simple, responsive client-side interface. Importance: High

2	Backend / Server-Side	Node.js + Express.js	Used to build the RESTful API and handle authentication, study-material management, AI requests, and communication between the frontend, database, and Gemini AI. Importance: High
3	Database / Data Storage	MongoDB	Stores user accounts, authentication data, uploaded study-material information, summaries, flashcards, quizzes, and study plans. Suitable for flexible document-based data. Importance: High
4	Cloud / Hosting / Deployment	Render	Provides simple cloud hosting for the Node.js and Express.js backend API with an easy deployment workflow suitable for a student project prototype. Importance: Medium
5	Version Control & CI/CD	Git + GitHub	Git provides version control and change tracking. GitHub stores source code, supports team collaboration, and maintains the project repository. Importance: High

6	Third-Party APIs / Other Tools	Google Gemini API + Postman + MongoDB Compass	Google Gemini API enables AI summaries, flashcards, quizzes, and personalized study plans. Postman tests REST endpoints, while MongoDB Compass helps view and manage MongoDB data during development. Importance: High